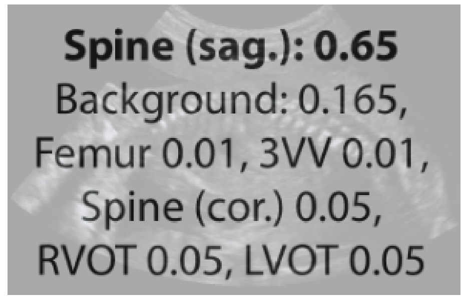
Use Case 5: Classification of diagnostic standard planes in ultrasound imaging



You are part of a research team developing an AI model for automatic classification of diagnostic standard planes in ultrasound imaging. The system is intended to support clinicians during image acquisition by identifying whether the currently acquired view corresponds to a clinically relevant standard plane.

The model analyzes ultrasound frames and predicts the most likely image category together with a confidence score for each possible class. A frame is assigned to a specific class once the predicted confidence exceeds a predefined decision threshold. However, clinicians are also interested in the confidence scores of alternative standard plane categories, particularly in ambiguous acquisition situations where several anatomically similar views may be plausible.

Some standard planes are considered substantially more clinically relevant than others because they are required for diagnostic assessment and reporting. In addition, the frequency of different image categories in the collected dataset does not fully reflect the frequency encountered during routine clinical examinations (prevalence).

Your dataset contains a large number of non-standard or background views acquired during free-hand ultrasound scanning, whereas some diagnostically important standard planes occur relatively infrequently. Nevertheless, the team does not explicitly want to compensate for this imbalance during metric selection, since the observed distribution is considered part of the realistic acquisition process.

Clinicians are interested not only in whether predictions are correct, but also in whether the predicted confidence scores are reliable and interpretable in isolation.

You now want to validate the model appropriately and decide which metrics should be used for reporting model performance.

Task description

Study your use case carefully. Then go to: https://metrics-reloaded.dkfz.de/metric-selection

Follow the decision process of the Metrics Reloaded toolkit and answer the fingerprint questions based on the information provided in the given scenario. Your goal is to identify the most appropriate performance metrics for the given application.

After finishing the task, save the generated PDF report containing the selected metrics and fingerprint answers. Be prepared to briefly present your use case to the group, including:

- how you interpreted the scenario,
- which fingerprint questions were easy or difficult to answer,
- whether any design choices were unclear or ambiguous,
- and which metrics were ultimately selected by the toolkit.

Please send your PDF report to the lecturers after completing the exercise.